

Eclipse Setup - Banner Admin Pages

Quick Guide

See this page [Banner Transformation - Clone Transformation Repository](#) for cloning details.

Pre-Requisite:

You Should have the following installed

- Maven
- Eclipse
- Tomcat - 7.0.47 recommended
- JDK -1.8 for your page development.

-1.8 You may also need a JDK 7 version to compile java batch. We still deliver only JDK 7 java batch jars.

Note/hint - If you are facing consistent issues with latest version of eclipse. install Eclipse 2019-12r (consider to use latest version if you don't have performance issues or other errors)

Maven Settings.xml file

Before you start you need to make sure your maven settings.xml file has the proper settings so that your eclipse can download Morphis plugins and other files. Since both Banner Admin and BAM utilize Maven, we only need to do this once.

- To edit your maven Settings, open the [conf/Settings.xml](#) file in maven or [.m2](#) folder and add the below sections

Mac users the .m2 folder is located in your home or root folder (/Users/username/.m2). Windows the .m2 folder is in your User folder.

Here is a sample, you ONLY need to add the mirrors and servers sections.

```

<?xml version="1.0" encoding="UTF-8"?>
<settings xsi:schemaLocation=" https://maven.apache.org/SETTINGS/1.1.0
https://maven.apache.org/xsd/settings-1.1.0.xsd " xmlns=" https://maven.apache.org/SETTINGS/1.1.0 "
  xmlns:xsi=" https://www.w3.org/2001/XMLSchema-instance ">

  <!-- pluginGroups
    | This is a list of additional group identifiers that will be searched when resolving
    plugins by their prefix, i.e.
    | when invoking a command line like "mvn prefix:goal". Maven will automatically add
    the group identifiers
    | "org.apache.maven.plugins" and "org.codehaus.mojo" if these are not already
    contained in the list.
    |-->
  <pluginGroups>
    <!-- pluginGroup
      | Specifies a further group identifier to use for plugin lookup.
      <pluginGroup>com.your.plugins</pluginGroup>
    -->

    <!-- jboss.org config start -->
    <pluginGroup>org.jboss.maven.plugins</pluginGroup>
    <!-- jboss.org config end -->

  </pluginGroups>

```

```

<mirrors>
<mirror>
  <!--This sends everything else to /public-->
  <id>nexus</id>
  <mirrorOf>*</mirrorOf>
  <url>http:// esmsvc02.ellucian.com :8181/nexus/content/groups/public</url>
</mirror>
</mirrors>

```

```

<servers>
  <server>
    <id>ellucian</id>
    <username>ellucian</username>
    <password>ellucian2014</password>
  </server>
</servers>

```

</settings>

Morphis Foundation Plugins

We use morphis plugin in eclipse and it can be added by going to *help>Install new software* on eclipse. and adding the following URL:

If you are installing Neon version of Eclipse install the following :

<http://downloads.cdn.morphis-tech.com/eclipse/release/mars/>

(OR)

Install the following for templates : <http://downloads.morphis-tech.com/eclipse/ellucian/release/4x>

For INTERNAL use only, here are the credentials you may need the first time you download the plugins

- User: ellucianuser
- Password: wW+E6?\$A

Make sure you accept all software updates. Foundations and Frames now utilizes a 3rd party software called Sapphire for storing preferences.

Current Version released to clients March 2019

Foundations Plugin 2.13.0

Frames 2.8.7

Templates Plugin 1.5

Version in test and with newer configurations

😊👍 Safe to start using now with Eclipse 2019-12

Foundations Plugin 2.19

Frames 2.8.8

Templates Plugin 1.7

Note: foundations 2.19 is required if you are using the 2019-12 and after of eclipse.

Global workspace settings

The following setup applies the Morphis plugin configuration to every workspace created. When these steps are followed, you will see, the foundation plugin, generator class preferences retained for every workspace and also it retains the preferences for the configuration section *too*. (*This was tested in Mars 2*)

- Locate the **config.ini** within **eclipse/configuration** folder and add this line to the bottom of the file.

For Mac users: Eclipse Mars on the Mac has been reworked to be a proper Mac application with everything included in the Eclipse.app package. You can find the config.ini in
/Applications/Eclipse.app/Contents/Eclipse/configuration/ directory

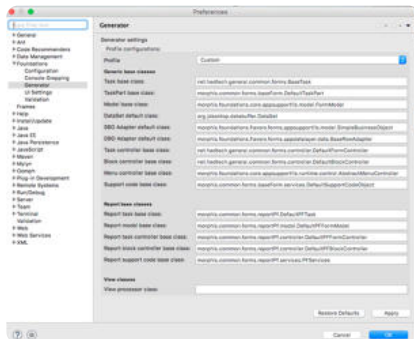
```
eclipse.pluginCustomization=<path>/morphis_eclipse.ini
```

morphis_eclipse.ini
29 Jan 2025, 04:54 AM

place the above file in the location you specified above. When you restart the eclipse, you should see the following configuration in your eclipse preferences from the next section.

• Foundation Generator

Access via the Window/Preferences menu.



Start [adding filters](#) in the filter tools menu.

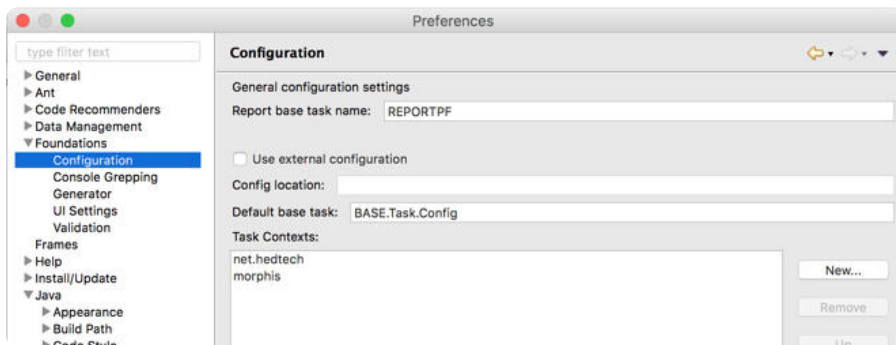
Task base class	net.hedtech.general.common.forms.BaseTask
TaskPart base class	morphis.common.forms.baseForm.DefaultTaskPart
Model base class	morphis.foundations.core.appsupportlib.model.FormModel
Dataset Default class	org.jdesktop.databuffer.DataSet
SBO Adapter Default Class	morphis.foundations.flavors.forms.appsupportlib.model.SimpleBusinessObject
DBO Adapter default class	morphis.foundations.flavors.forms.appdatalayer.data.BaseRowAdapter
Task controller base class	net.hedtech.general.common.forms.controller.DefaultFormController
Block controller base class	net.hedtech.general.common.forms.controller.DefaultBlockController
Menu controller base class	morphis.foundations.core.appsupportlib.runtime.control.AbstractMenuController
Support code base class	morphis.common.forms.baseForm.services.DefaultSupportCodeObject

Report task base class	morphis.common.forms.reportPf.DefaultPFTask
Report model base class	morphis.common.forms.reportPf.model.DefaultPFFormModel
Report task controller base class	morphis.common.forms.reportPf.controller.DefaultPFFormController
Report block controller base class	morphis.common.forms.reportPf.controller.DefaultPFBlockController
Report support code base class	morphis.common.forms.reportPf.services.PFServices

You can leave the View Classes blank.

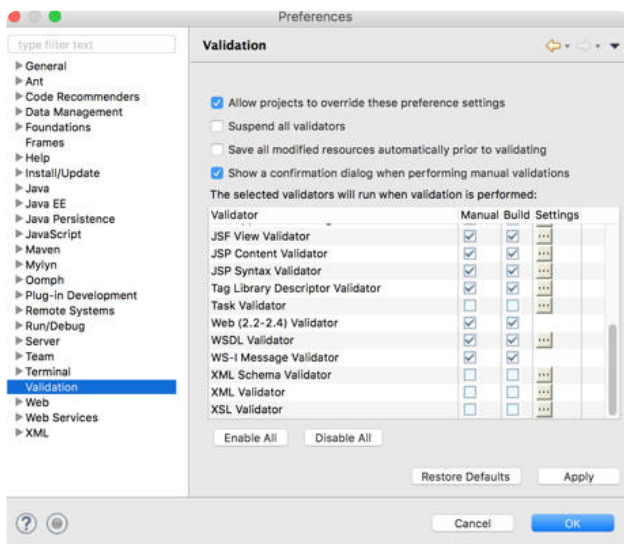
• Foundations Configuration

Click on the foundations configuration and add net.hedtech and also change the default base task to BASE.Task.Config.



• XML Validation

Make sure you uncheck the bottom 3 XML and XSL validator in the Eclipse / Preferences / Validation as shown here



Sizing

Setting the memory size and other related parameters are important so your app runs fast enough.

JDK Sizing

Basic sizing for JDK7 is:

```
-XX:MaxPermSize=512m -XX:PermSize=512m -Xms1g -Xmx4g -server -  
XX:+CMSClassUnloadingEnabled -XX:-UseGCOverheadLimit -  
DPROXY_SERVER_NAME=http:// esmsvc02.ellucian.com :8181/nexus/content/groups/public -  
Dhttp.proxyPort=8080
```

Basic sizing for JDK8 is:

```
-Xms1g -Xmx4g -server -  
DPROXY_SERVER_NAME=http:// esmsvc02.ellucian.com :8181/nexus/content/groups/public -  
Dhttp.proxyPort=8080
```

Make sure you remove ' -Dhttp.proxyHost=www-proxy.sct.com' the proxy is no longer valid for our vpn

Eclipse Sizing

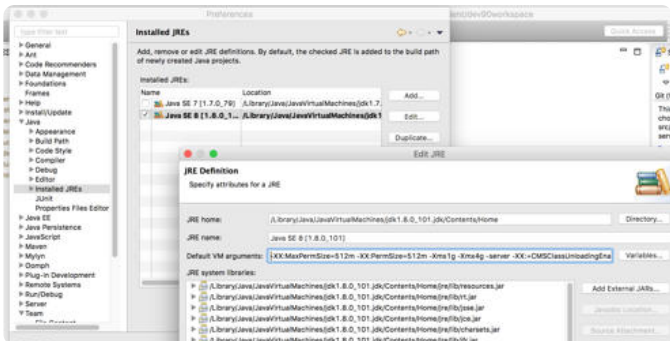
Follow instructions to increase the java heap size for your app by adding the xms, xmx to your eclipse.ini
<http://crunchify.com/increase-eclipse-memory-size-to-avoid-oom-on-startup/>

Environment Variables

In windows add the sizing to your JAVA_OPTS environment variable, and on MAC to the java_ops in your .bash_profile file.

Eclipse Heap Size

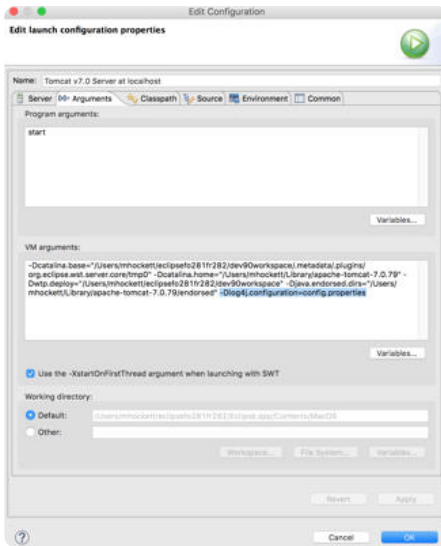
Open the eclipse properties / Java / Installed JREs. Check your JRE and press the Edit Button. Now enter the space variables in the VM arguments.



Run Time Configuration

Add the parameter for the configuration at the end of the VM arguments in the Run / Run Configuration.

```
-Dlog4j.configuration=config.properties
```



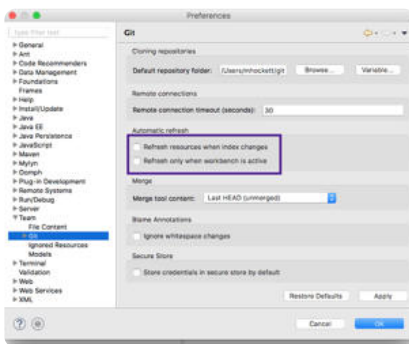
add the parameter `-Djava.io.tmpdir=/Users/mhockett/tempupload` to specify where upload files are saved.

Team - GIT Indexing

Turn off automatic GIT indexing in your eclipse application.

Open Preferences / Team/ Git

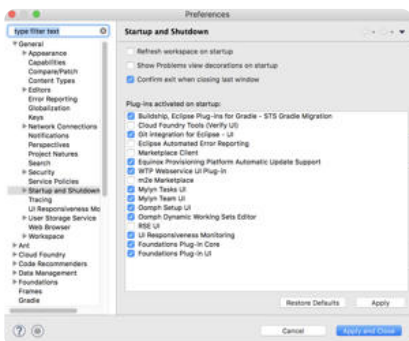
Uncheck the two checkboxes for automatic refresh.



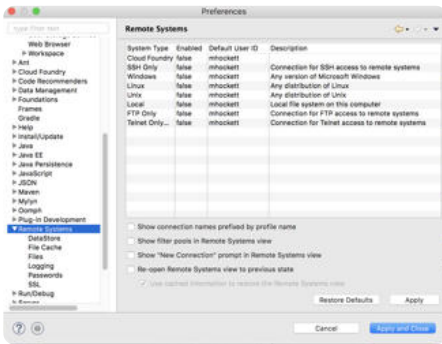
Other tips to speed up your eclipse

Remove some of the start up processes, like the shown here. Open preferences and type startup in the filter.

uncheck RSE UI and the others as they are not used for admin page development

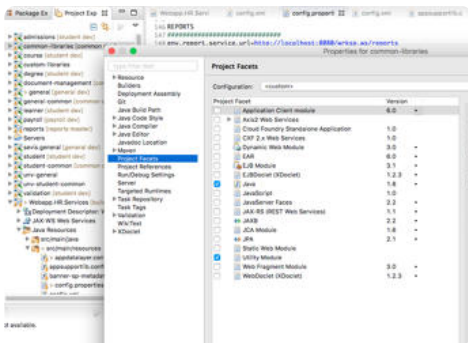


removing remote systems also helps

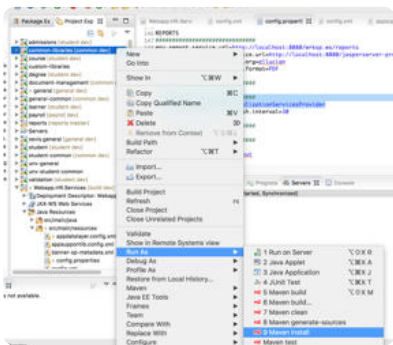


Troubleshoot

Make sure that these are set for your project. Right click on project, select properties, and check the two items as shown.



You can try to build common stand alone



Required GIT Repos and Imports

Banner Admin

Start your development project by creating a main directory `ban_admin_pages`, with two sub directories `dev90` and `dev90workspace`.

Clone your projects in to dev90, and open the ban_admin_ages/dev90workspace as your eclipse workspace

Copy the following clone shell script into your `ban_admin_pages` directory and execute.

The following example clone shell script brings in the dev branch. Please review [Transformation GIT Branch Tracking#Overall](#) for current branch versions.

The following projects are required to build your app, and must be imported into your workspace

`common-libraries`

`document-management`

`general-common`

`reports`

`student-common`

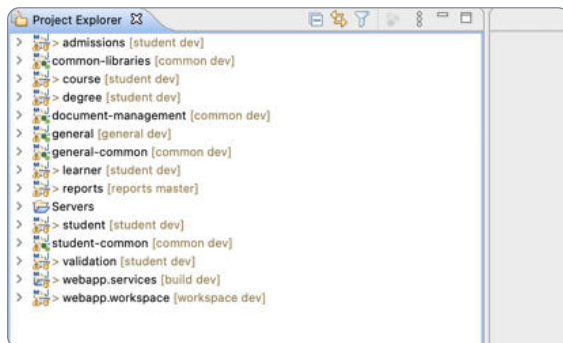
[webapp.hr](#).workspace (workspace)

[webapp.hr](#).services (build)

Banner Admin directory structure

After creating the directory structure under `ban_admin_pages` and cloning Banner Admin, you should end up with:

```
ban_admin_pages
|-- dev90
|   |-- base
|   |-- build
|   |-- common
|   |-- general
|   |-- reports
|   \-- workspace
\-- dev90workspace
```



Tomcat

You can run using tomcat 7 or 8, please be advised on bug with tomcat 8

<https://ellucian.force.com/clients/articles/FAQ/Cache-issues-when-using-Tomcat-8-x-in-an-Eclipse-development-environment>

Basically you may need to uncheck the serve modules without publishing if you are using tomcat 8, and then before each run you need to right click on the server and clean, and clean tomcat working directory and then publish.

- ☐ Copy the jar files in the .m2/repositories/com/oracle/ojdbc7/12.1.10.1/ojdbc7-12.1.0.1.jar and .m2/repositories/com/oracle/ucp/11.2.0.4/ucp-11.2.0.4.jar to your tomcat/lib folder.

Speed up Tomcat startup

- a. Open the two files:

```
webapp-services/src/main/config/WEB-INF/web.xml
webapp-services/src/main/webapp/WEB-INF/web.xml
```

2. Modify the top level xml attribute, add metadata-complete="true":

- a. Before:

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app id="WebApp_Wrksp_WS_ID" version="2.5"
  xmlns="http://java.sun.com/xml/ns/javaee" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
  xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd">
```

- b. After

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app id="WebApp_Wrksp_WS_ID" version="2.5"
  xmlns="http://java.sun.com/xml/ns/javaee" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
  xsi:schemaLocation="http://java.sun.com/xml/ns/javaee
http://java.sun.com/xml/ns/javaee/web-app_2_5.xsd"
  metadata-complete="true">
```

Quick Build(troubleshooting)

1. Getting error during the maven update and can't determine the issue:

```
try these:
-- delete the morphis directory from the .m2/repository
cd into dev90/workspace/webapp-workspace    # For Banner Admin
mvn clean install -U
-- if you get an error and you want to see it, try
mvn clean install -U -e
```

2. Quick clean instead of using Project / Clean

This is the same above but without the -U parameter

```
cd dev90/workspace/webapp-workspace    # For Banner Admin
mvn clean install
```

3. Maven Update at command line

```
cd dev90/build/ net.hedtech.banner.hr    # For Banner Admin
-- make sure pom.xml has all your projects and is correct
mvn package
```

POST SET-UP

Follow these steps after setup to run the pages:

1. Adding required dependencies -

- In project explorer */webapp.services/pom.xml* uncomment the required dependencies. For etranscripts/Banner Student we need the following dependencies

```

<dependency>
  <groupId>net.hedtech.banner.general</groupId>
  <artifactId>general</artifactId>
  <version>${project.parent.version}</version>
</dependency>

<dependency>
  <groupId>net.hedtech.banner.student</groupId>
  <artifactId>admissions</artifactId>
  <version>${project.parent.version}</version>
</dependency>

<dependency>
  <groupId>net.hedtech.banner.student</groupId>
  <artifactId>course</artifactId>
  <version>${project.parent.version}</version>
</dependency>

<dependency>
  <groupId>net.hedtech.banner.student</groupId>
  <artifactId>degree</artifactId>
  <version>${project.parent.version}</version>
</dependency>

<dependency>
  <groupId>net.hedtech.banner.student</groupId>
  <artifactId>learner</artifactId>
  <version>${project.parent.version}</version>
</dependency>

<dependency>
  <groupId>net.hedtech.banner.student</groupId>
  <artifactId>validation</artifactId>
  <version>${project.parent.version}</version>
</dependency>

<dependency>
  <groupId>net.hedtech.banner.student</groupId>
  <artifactId>student</artifactId>
  <version>${project.parent.version}</version>
</dependency>

```

2. In project explorer open the following `/webapp.services/src/main/resources` and edit the following configuration files
 - In **config.properties** - edit the database configuration and add your database server's IP, username, password

```
connection.user = <username>

connection.password = <password>

connection.host = <database>:1521

connection.database = GVUDB
```

- In **appdatalayer.config** – search and uncomment the following line

```
<entry key="PKG_XML" value="goktxml"/>
```

3. Change your tomcat settings by double clicking on the added server

- Under server options, **enable serve modules without publishing**
- Under Timeouts, **increase the start time and stop time to 450 and 150**

Setting Trace to your Session

Launch tomcat with start_trace

Application

Banner Admin	http://localhost:8080/wrksp/	http://localhost:8080/wrksp/?START_TRACE=YES
--------------	---	---